

Basis of difference	Public	Private	Consortium
Accessibility	Anyone	Single user	More than one central in-charge
Who can join?	Anyone	Permissioned or known identities	Permissioned or known identities
Consensus mechanism	PoS/PoW	Voting	Voting
Speed of transaction	Slow	Faster	Faster
Decentralisation	Completely and highly decentralised	Distributed	Decentralised
Transactions reading	Any node	Any node (without permission) or a list of predefined nodes	Any node or a list of predefined nodes
Data immutability	Yes, blockchain rollback is almost impossible	Yes, but blockchain rollback is possible	Yes, but blockchain rollback is possible
Features	• Censorship resistance • Unregulated and cross-border use • Support for native assets • Anonymous identities • Scalable network architecture	• Applicable to highly regulated businesses • Efficient transactions throughput • Transactions without fees • Infrastructure rules are easier to manage • Better protection against external disturbances	
Examples	• Bitcoin, Ethereum, Ripple, etc.	• MultiChain, Quorum, Hyperledger, Tendermint, etc.	

Table 1: Differences between public, private and consortium blockchains

Advantages of a blockchain consortium

- **Cost savings:** The basic idea of being a member of a blockchain consortium is to cut expenses. It helps businesses cut costs.
- **Helps to accelerate learning:** Blockchain consortia offer strong platforms for learning in terms of training and development support from experts.
- **Risk sharing:** As blockchain is a relatively new technology, its adoption comes with a few risks. In a consortium, members collaborate with each other and thus reduce the risk factor, irrespective of whether the final solutions fail or succeed.
- **Mass adoption:** Consortia help in mass adoption of the technology.

- **Lifetime maintenance:** A consortium offers a technology roadmap, helping businesses grow by leaps and bounds.
- **Influencing standards:** Blockchain consortia are set to influence standards as blockchain will soon reach interoperability. The blockchain industry is expected to become a global trillion-dollar industry.

Governance model for blockchain consortium

Blockchain consortia have become popular in the past few years and are widely replacing legacy distributed workflow systems like simplified document management or semi-manual compliance systems.

Typically, in a blockchain consortium, we have various key entities, as shown in Figure 2 and explained below.

Founding member, who develops the governance model and the hierarchy of groups to be used in the blockchain platform.

Consortium promoter, who performs the administrative activity for implementing the blockchain platform of choice (R3, Hyperledger, B3i, Bankchain, etc), depending on the industrial solution and use case being addressed.

Participating members, who are the actual users of the blockchain platform; they handle the transactions and prepare the interconnection between the nodes in the blockchain platform.

Governance principles define the principles to be used in the public and